

LAB PROJECT REPORT

Course Title: CSE422 - Artificial Intelligence

**Project Title: Comprehensive Empirical Evaluation of Supervised
Classification Models and K-Means Clustering on Financial Risk Dataset 14**

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Comprehensive Empirical Evaluation of Supervised Classification Models and K-Means Clustering on Financial Risk Dataset 14

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Abstract—In this project, the target feature selected for predictive modeling is `loan_status`, which represents binary discrete values (0 for non-default and 1 for default). Financial institutions face significant challenges in assessing loan default risks accurately. The objective of this project is to apply supervised classification models and unsupervised clustering to predict applicant defaults using financial and demographic indicators. With the help of machine learning pipelines, the goal is to reduce non-performing loans and automate risk management decision-making.

Index Terms—Supervised Learning, K-Means Clustering, Classification, Feature Scaling, Machine Learning.

I. INTRODUCTION

MACHINE learning algorithms are fundamental to predictive modeling and pattern recognition in structured datasets. In this project, the target feature selected for predictive modeling is `loan_status`, which represents binary discrete values (0 for non-default and 1 for default). Financial institutions face significant challenges in assessing loan default risks accurately. The objective of this project is to apply supervised classification models and unsupervised clustering to predict applicant defaults using financial and demographic indicators. By leveraging machine learning pipelines, the goal is to reduce non-performing loans and automate risk management decision-making.

II. DATASET DESCRIPTION

- **Problem Type:** This is a Classification Problem because the target feature `loan_status` contains discrete binary outcomes (0 and 1) rather than a continuous numeric range.
- **Dataset Size & Features:** The dataset consists of 45,000 applicant records and 12 distinct feature columns.
- **Types of Features:**
 - *Quantitative (Numerical):* `person_age`, `person_income`, `person_emp_length`, `loan_amnt`, `loan_int_rate`, `loan_percent_income`, `cb_person_cred_hist_length`.
 - *Categorical:* `person_home_ownership`, `loan_intent`, `loan_grade`, `cb_person_default_on_file`.
- **Dataset Balance / Imbalance:** The target feature is imbalanced. The unique classes do not have equal instances; Class 0 accounts for approximately 77.78% (35,000

instances) while Class 1 accounts for 22.22% (10,000 instances).

- **Class Distribution Representation:** Class 0 (Non-Default): 35,000 applicants (77.78%), Class 1 (Default): 10,000 applicants (22.22%) (Represented via a 2-class Bar Chart in the notebook analysis).

III. DATA ANALYSIS

Exploratory Data Analysis (EDA) was conducted to extract statistical summaries and correlation structures. Correlation matrix analysis revealed a strong positive correlation between `loan_percent_income` and `loan_status`, indicating that high debt-to-income ratios directly increase default likelihood. Furthermore, continuous attributes like `person_income` showed negative correlation with loan default, confirming higher income stability reduces default probability.

IV. DATASET PRE-PROCESSING

- **Handling Null and Duplicate Values:** Missing entries existed in continuous columns such as `loan_int_rate` and `person_emp_length`. Implemented median strategy imputation (`SimpleImputer`) for numerical columns and mode imputation for categorical columns to avoid dropping valuable rows.
- **Encoding Categorical Values:** Machine learning algorithms require numerical input formats. One-Hot Encoding (`pd.get_dummies`) was applied across categorical features (`person_home_ownership`, `loan_intent`, `loan_grade`).
- **Feature Selection & Leakage Prevention:** Target variable leakage during training was solved by isolating `loan_status` completely from the predictive feature matrix X prior to model execution.
- **Feature Scaling:** Varying scales across features skew distance-based estimators. Applied `StandardScaler` to scale features to zero mean and unit variance.
- **Imbalance Handling:** Applied SMOTE (Synthetic Minority Over-sampling Technique) to address class imbalance in the training split.

V. DATASET SPLITTING

- **Strategy:** Stratified Train-Test Splitting (`stratify=y`) was implemented to preserve class proportions in both sets.

- **Train Set:** 80% of total data (36,000 samples).
- **Test Set:** 20% of total data (9,000 samples).

VI. MODEL TRAINING & TESTING (SUPERVISED)

Four distinct classification models were instantiated and evaluated:

- **K-Nearest Neighbors (KNN):** Trained using feature-space distance calculations.
- **Logistic Regression:** Trained as a linear baseline classification model.
- **Naive Bayes:** Applied assuming feature independence.
- **Neural Network (MLP/TensorFlow/PyTorch):** Configured to capture non-linear relationships across iteration epochs.

VII. MODEL SELECTION / COMPARISON ANALYSIS

- **Accuracy Comparison:** Logistic Regression and Neural Networks exhibited the highest overall prediction consistency across evaluation sets.
- **Precision, Recall, & F1-Score:** Neural Network achieved balanced F1-scores on the minority class (1), whereas Naive Bayes showed higher recall at the expense of precision.
- **Confusion Matrix Analysis:** Confirmed that applying SMOTE reduced false negative rates significantly in Logistic Regression and Neural Network outputs.
- **AUC Score & ROC Curve:** Logistic Regression achieved the most stable Area Under Curve (AUC) performance metrics across standard threshold variations.

VIII. UNSUPERVISED LEARNING

Removed the target label `loan_status` to evaluate unlabelled clustering performance. K-Means Clustering was executed on key features (`person_age` and `person_income`). The Elbow Method ($k = 1$ to 9) was applied to compute Within-Cluster Sum of Squares (WCSS) inertia, establishing an optimal cluster configuration at $k = 3$ with clear centroid boundaries.

IX. CONCLUSION

- **Best Model:** Logistic Regression (alongside Neural Network) achieved the highest overall consistency and predictive stability on dataset 14.
- **Reasons for Performance Variations:** Unscaled features negatively impacted distance-based estimators like KNN and K-Means prior to standardization.
- **Challenges Faced:** Key challenges included handling target class imbalance, performing leak-free categorical encoding, and calibrating neural network hyper-parameters.